

Chapter 1

Section 4 - Number Theory

1.1 Section 4.1 - Divisibility

Question

When is an integer divisible by 2?

Solution:

When an integer ends in 0, 2, 4, 6, 8

Let's prove this after we develop some notation

Definition 1 (Divides)

For integers $a, b, \in \mathbb{Z}$, we say "a divides b" if b is divisibly by a and we write

$$a \mid b \quad (\text{TRUE value})$$

$$a \nmid b \quad (\text{FALSE value})$$

Example 1

$2 \mid 48$ since 48 is divisible by 2

but $7 \nmid 48$ since 48 is not divisible by 7

Let's prove

for $n \in \mathbb{Z}, 2 \mid n$ if $\exists k \in \mathbb{Z}$ such that $n = 10k$ or $10k + 2$ or $10k + 4$ or $10k + 6$ or $10k + 8$

We could also say

$$\exists k \in \mathbb{Z} \text{ such that } n = 10k + r \text{ where} \\ r = \{0, 2, 4, 6, 8\}$$

Proof

(Proof of the converse)

Let $n \in \text{set } Z$

Then base 10 representation of n is of the form

$$n = d_k 10^k + d_{k-1} 10^{k-1} + \dots + d_1 10 + d_0$$

Since $2 \mid n$, then $n = 2m$ for some $m \in \mathbb{Z}$

We have

$$n = 10(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) + d_0 \\ n = 2 \left[5(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) \right] + d_0$$

Solving for d_0 we get

$$d_0 = n - 2 \left[5(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) \right]$$

Since d_0 is the sum of two even numbers, then d_0 is even and $d_0 \in \{0, 2, 4, 6, 8\}$

Let's prove some of the properties of the "divides" symbol

In particular, let's rewrite what $a \mid b$ means for $a, b \in \mathbb{Z}$

So, $a \mid b$ means

$$\exists k \in \mathbb{Z} \text{ such that } b = ak$$

We believe if two integers are divisible by an integer, then their sum is divisible by that integer

We prove $a, b, c \in \mathbb{Z}, a \mid b, a \mid c, \implies a \mid (b + c)$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$a \mid c \implies \exists n \in \mathbb{Z} \text{ such that } c = an$$

Then

$$b + c = ak + an = a(k + n)$$

$$\therefore a \mid (b + c) \text{ as desired}$$

Note

This also works for **subtraction**

Proposition 1 (Modulus Multiplication)

$$a, b, c \in \mathbb{Z}, a \mid b, a \mid c \implies a^2 \mid bc$$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$a \mid c \implies \exists n \in \mathbb{Z} \text{ such that } c = an$$

Then

$$bc = (ak)(an)$$

$$= a^2(kn), \text{ proving } a^2 \mid bc \text{ as desired}$$

Proposition 2 (Constant Factor)

$$a, b, c \in \mathbb{Z}, a \mid b \implies a \mid bc$$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$bc = akc = a(kc), \text{ implying } a \mid bc, \text{ as desired}$$

Let's rewrite the old proof from earlier, this time proving the actual proposition rather than contrapositive

Proposition 3

For $n \in \mathbb{Z}$, where $n = 10k + r, r \in \{0, 2, 4, 6, 8\}$, then $2 \mid n$

Proof

$$\forall r \in \{0, 2, 4, 6, 8\}, \exists m \in \mathbb{Z} \text{ such that } r = 2m$$

Subbing this in

$$\begin{aligned} n &= 10k + r = 10k + 2m \\ &= 2(5k + m), \text{ implying } 2 \mid n, \text{ as desired} \end{aligned}$$

Proposition 4

For $n \in \mathbb{Z}$, if the sum of the digits is divisible by 3, then the integer n is divisible by 3

Proof.

$$\begin{aligned} n &= d_k 10^k + d_{k-1} 10^{k-1} + \dots + d_1 10 + d_0 \\ n &= (10^k - 1)d_k + (10^{k-1} - 1)d_{k-1} + \dots + 99d_2 + 9d_1 + \underbrace{(d_k + d_{k-1} + \dots + d_2 + d_1 + d_0)}_{\text{sum of digits}} \end{aligned}$$

Because 10^k and so on will always be a bunch of 9's, we see that

$$3 \mid (10^k - 1)d_k + (10^{k-1} - 1)d_{k-1} + \dots + 99d_2 + 9d_1$$

And

$$3 \mid (d_k + d_{k-1} + \dots + d_2 + d_1 + d_0) \text{ by hypothesis}$$

$\therefore n$ is the sum of 2 multiples of 3, proving that $3 \mid n$ as desired

□

Here's just a table for divisibility rules

n is divisible by	if
2	last digit is even
3	sum of digits is divisible by 3
4	last 2 digits is divisible by 4
5	last digit is 0 or 5
6	divisible by both 2 and 3
7	He showed us but I forgot
8	last 3 digits divisible by 8
9	sum of digits is divisible by 9
10	last digit is 0

Definition 2 (Division Algorithm)

The **division algorithm** is a way of expressing division with integers

$$\forall a \in \mathbb{Z} \forall d \in \mathbb{Z}^+ \exists q \in \mathbb{Z} \exists r \in \{0, 1, 2, 3, \dots, d-1\}, a = dq + r$$

Where

d : divisor
 q : quotient
 r : remainder

Example 2

Let $a = 341$ and $d = 7$

$$341 = \underbrace{7}_{\text{divisor}} \cdot \underbrace{48}_{\text{quotient}} + \underbrace{5}_{\text{remainder}}$$

However we can write this in a more compact way

$$341 \pmod{\underbrace{7}_{\text{divisor}}} = \underbrace{5}_{\text{remainder}}$$

Example 3

Let's give examples of division, we're practically finding the remainder

1.

$$\frac{6568}{9} = 729.777$$
$$6568 - 9 \cdot 729 = 7$$

2.

$$\frac{343348}{7} = 49049.714$$
$$343348 - 7 \cdot 49049 = 5$$

3.

$$\frac{-100}{13} = -7.6923$$
$$-100 - 13(-7) = -9$$
$$= -100 - 13(-8) = 4$$
$$-100 = (13)(-8) + 4$$

Note

For negatives, round down to the next negative integer in order to get a positive remainder

Example 4

Let's do more examples but with mod

1.

$$1023 \pmod{11} = 0$$

2.

$$6568 \pmod{9} = 7$$

3.

$$343348 \pmod{7} = 5$$

4.

$$-100 \pmod{13} = 4$$

Definition 3 (Congruence)

Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$

We say a is **congruent** to b modulo m if

$$a \equiv b \pmod{m}$$

This is basically saying are the remainders the same

Written as

$$a \equiv b \pmod{m}$$

And if they're not then

$$a \not\equiv b \pmod{m}$$

Note

If $a \equiv b \pmod{m}$ then by **Division Algorithm**

$$a = mq_a + r \text{ and } b = mq_b + r$$

Subtrating the equations we get

$$\begin{aligned} a - b &= (mq_a + r) - (mq_b + r) \\ &= m(q_a - q_b) \end{aligned}$$

$$\boxed{\therefore m \mid (a - b)}$$

Proposition 5

Let $a, b, c, d \in \mathbb{Z}, m \in \mathbb{Z}^+$

Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$

1. $a + c \equiv b + d \pmod{m}$

Proof

$$a \equiv b \pmod{m} \implies m \mid (a - b)$$

$$c \equiv d \pmod{m} \implies m \mid (c - d)$$

Then

$$\begin{aligned}m &\mid [(a-b) + (c-d)] \\m &\mid [(a+c) - (b+d)] \\ \boxed{a+c \equiv b+d \pmod{m}} &\text{ as desired}\end{aligned}$$

2. $ac \equiv bd \pmod{m}$

Proof

$$\begin{aligned}a \equiv b \pmod{m} &\implies m \mid (a-b) \\c \equiv d \pmod{m} &\implies m \mid (c-d)\end{aligned}$$

$$\begin{aligned}m \mid (a-b) &\implies \exists k_1 \in \mathbb{Z} \text{ such that } a-b = mk_1 \\m \mid (c-d) &\implies \exists k_2 \in \mathbb{Z} \text{ such that } c-d = mk_2\end{aligned}$$

Then $a = b + mk_1$ and $c = d + mk_2$

$$\begin{aligned}ac &= (b + mk_1)(d + mk_2) && \text{(FOIL)} \\&= bd + bmk_2 + dm k_1 + m^2 k_1 k_2\end{aligned}$$

Moving bd

$$\begin{aligned}ac - bd &= bmk_2 + dm k_1 + m^2 k_1 k_2 \\&= m(bk_2 + dk_1 + mk_1 k_2) \\ \boxed{\therefore m \mid (ac - bd)}\end{aligned}$$

3. if $h \in \mathbb{Z}$ then $ah \equiv bh \pmod{m}$

Proof

$$a \equiv b \pmod{m} \implies m \mid (a - b)$$

Then since $h \in \mathbb{Z}$, we have

$$m \mid h(a - b)$$

$$m \mid (ah - bh)$$

$$\boxed{ah \equiv bh \pmod{m}}$$

1.2 Section 4.2 - Binary

1.3 Section 4.3 - Primes

Definition 4

A **prime number** is a positive integer with exactly 2 **distinct factors**

Example 5

Here are the first few primes

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ...

is 257 prime?

We use **trial division**

We'll use a loop

```
for k = 2 to 256
  if 257 % k = 0
    then return "composite"
  else
    k += 1
```

For efficiency, we could just go up to $\sqrt{257}$, or $\lfloor \sqrt{256} \rfloor$

Second method: **Sieve of Eratosthenes**

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

I honestly forgot how this works since I didn't write it down too well

Third method: **Fermat's Method**

if $n = a^2 - b^2$

then $n = (a - b)(a + b)$

Check: $a - b > 1$ and $a + b > 1$

If neither is 1, then n is composite. If both are greater than 1, then they can't be n

Fourth Method: **Quadratic Sieve**

Solve

$$a^2 - b^2 \equiv 0 \pmod{n}$$

Then

$$n \mid a^2 - b^2 \implies a^2 - b^2 = nk, k \in \mathbb{Z}$$

Proposition 6

There are infinitely primes

Proof

Suppose on the contrary, there are finitely many primes such that all primes can be written like

$$p_1, p_2, \dots, p_k$$

Consider

$$N = p_1 p_2 \cdots p_k + 1$$

Case 1: Suppose N is prime

That means that

N is larger than p_i for $i = 1, 2, \dots, k$

Hence $N \neq p_i$ for $i = 1, 2, \dots, k$. We have a contradiction of our finitely many primes!

Case 2: N is composite

Then N must have a prime factor in the list $p_1, p_2, \dots, p_k$

Suppose p_j is a prime factor of N

Then $p_j \mid N$ and not $p_j \mid p_1 p_2 \cdots p_k$

Then we conclude

$$p_j \mid (N - p_1 p_2 \cdots p_k)$$

Therefore

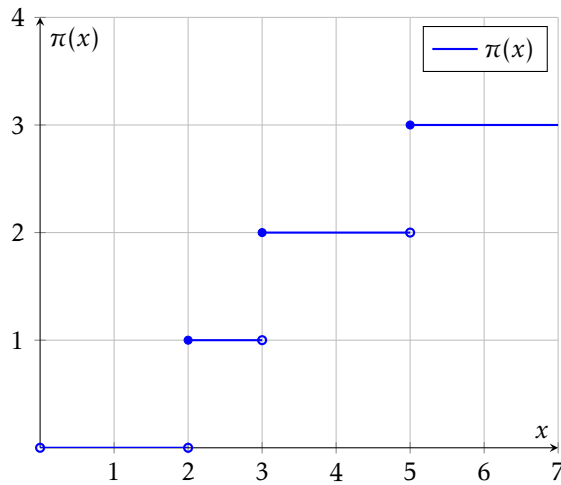
$$p_j \mid 1, \text{ is a contradiction}$$

We conclude there are infinitely primes, as desired (by contradiction)

Definition 5 (Prime Counting Function (Pi))

The function $\pi : \mathbb{R}^+ \rightarrow \mathbb{Z}^+$ is defined by

$$\pi(x) = \text{number of primes less than or equal to } x$$



Theorem 1 (The Prime Number Theorem)

This theorem is defined that $\pi(x)$ is approximately equal to $\frac{x}{\ln(x)}$ as $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\frac{x}{\ln(x)}} = 1$$

In other words,

$$\pi(x) \approx \frac{x}{\ln(x)}$$

Note

This was proven independently by Jacques Hadamard and Charles Jean de la Vallée Poussin in 1896

To recap, we know so far for $a, b, c, d \in \mathbb{Z}, m \in \mathbb{Z}^+$ that

1. $a \equiv b \pmod{m} \iff m \mid (a - b)$
2. $a \equiv b \pmod{m}, c \equiv d \pmod{m}$
 - (i) $a + c \equiv b + d \pmod{m}$
 - (ii) $ac \equiv bd \pmod{m}$
 - (iii) $ac \equiv bc \pmod{m}$

But we see that we can multiply and add with $\equiv$, but how can we divide with modular arithmetic?

Let's go back to algebra

How do we solve this? We need a way to get rid of the 7

$$\begin{aligned} 7x &= 14 \\ \frac{7}{7}x &= \frac{14}{7} \\ 1 \cdot x &= 2 \\ \boxed{x = 2} \end{aligned}$$

In this case, we used the **multiplicative inverse** of 7, 7^{-1} , which is $\frac{1}{7}$

$$\begin{aligned} 7x &= 14 \\ 7^{-1}7x &= 7^{-1} \cdot 14 && \text{(Multiplicative Inverse)} \\ 1x &= 2 && \text{(Multiplicative Identity)} \\ x &= 2 \end{aligned}$$

In order to divide, mod m , we need to understand **multiplicative inverse**

1 is still the multiplicative identity, so given $a \in \mathbb{Z}$, the multiplicative inverse of a (which we'll call x) satisfies

$$ax \equiv 1 \pmod{m}$$

We conclude that

$$m \mid (ax - 1)$$

This means that

$\exists k \in \mathbb{Z}$ such that

$$ax - 1 = mk$$

and

$$ax - mk = 1$$

Our first task is: when does this equation have a solution?

Claim 1

If $m \mid a$ then $ax - mk = 1$ has no integer solutions

Proof

If $m \mid a$ then $\exists n \in \mathbb{Z}$ such that $a = nm$

The equation $ax - mk = 1$ becomes

$$ax - mk = 1$$

$$nm - mk = 1$$

(Substitute)

$$m(n - k) = 1$$

(Factor m)

We conclude that $m \mid 1$, which is not possible

So $ax - mk = 1$ has no integer solutions

Example 6

Let's do an example

Let $m = 12, a = 7$

We want to find the multiplicative inverse of 7 modulo 12

We must solve

$$7x \equiv 1 \pmod{12}$$

Method 1: Brute Force

$$\begin{aligned}
7 \cdot 1 &\pmod{12} = 7 \\
7 \cdot 2 &\pmod{12} = 2 \\
7 \cdot 3 &\pmod{12} = 9 \\
7 \cdot 4 &\pmod{12} = 4 \\
7 \cdot 5 &\pmod{12} = 11 \\
7 \cdot 6 &\pmod{12} = 6 \\
7 \cdot 7 &\pmod{12} = 1
\end{aligned}$$

$$\boxed{7^{-1} \equiv 7 \pmod{12}}$$

This wasn't too long, but it can be very tedious and very long as we work with higher numbers. This works, but it's not good.

Method 2: Adding 0

Since $0 \equiv 12 \pmod{12}$, we can keep on adding 12 to the right side until it's divisible by 7

$$\begin{array}{r}
7x \equiv 1 \pmod{12} \\
+0 \equiv 48 \pmod{12} \\
\hline
7x \equiv 49 \pmod{12} \\
7^{-1}7x \equiv 7^{-1}7 \cdot 7 \pmod{12} \\
\boxed{x \equiv 7 \pmod{12}}
\end{array}$$

This was pretty short, but we got lucky. Normally this would take a lot more steps / additions

Method 3: Euclidean Algorithm

This is the one we'll be using most in this class

Let's rewrite

$$7x \equiv 1 \pmod{12}$$

We can rewrite it as divides

$$12 \mid (7x - 1)$$

Since 12 divides $(7x - 1)$, that means $(7x - 1)$ is a multiple of 12, so

$$\begin{aligned} 7x - 1 &= 12k \\ 7x - 12k &= 1 \end{aligned}$$

Now we just repeat the **division algorithm**

We start by breaking up 12

$$\begin{aligned} 12 &= 1 \cdot 7 + 5 \\ 7 &= 1 \cdot 5 + 2 \\ 5 &= 2 \cdot 2 + 1 \end{aligned}$$

Once we have the remainder as 1, we must work backwards

From the equations above, let's solve for each remainder

$$\begin{aligned} 5 &= 2 \cdot 2 + 1 \\ 1 &= 5 - 2 \cdot 2 \end{aligned} \quad \text{(Let's solve for 1)}$$

$$\begin{aligned} 7 &= 1 \cdot 5 + 2 \\ 2 &= 7 - 1 \cdot 5 \end{aligned} \quad \text{(Solve for 2)}$$

$$\begin{aligned} 12 &= 1 \cdot 7 + 5 \\ 5 &= 12 - 1 \cdot 7 \end{aligned} \quad \text{(Solve for 5)}$$

Now, using $1 = 5 - 2 \cdot 2$ as our base, let's repeatedly substitute the remainders

For each line, substitute the divisor then simplify

$$\begin{aligned}
1 &= 5 - 2 \cdot 2 && \text{(now substitute 2)} \\
1 &= 5 - 2(7 - 1 \cdot 5) && \text{(now simplify)} \\
1 &= 5 - 2 \cdot 7 + 2 \cdot 5 \\
1 &= 3 \cdot 5 - 2 \cdot 7 && \text{(now substitute 5)} \\
1 &= 3(12 - 1 \cdot 7) - 2 \cdot 7 && \text{(now simplify)} \\
1 &= 3 \cdot 12 - 3 \cdot 7 - 2 \cdot 7 \\
1 &= 3 \cdot 12 - 5 \cdot 7
\end{aligned}$$

Remember that we were trying to solve for $7x - 12k = 1$, so we conclude that

$$\begin{aligned}
7x - 12k &= 1 \\
7(-5) - 12(-3) &= 1 \\
x = -5, k = 3
\end{aligned}$$

We conclude that

$$\begin{aligned}
x &\equiv -5 \pmod{12} \\
+0 &\equiv 12 \pmod{12}
\end{aligned}$$

$$x \equiv 7 \pmod{12}$$

1.3.1 Euclidean Algorithm

Definition 6 (Euclidean Algorithm)

Because that last example could be hard to read, I'll rewrite the Euclidean Algorithm but with variables

Let's say we're trying to find the multiplicative inverse of

$$a \pmod{m}$$

We essentially solve

$$a \cdot a^{-1} - mk = 1, \text{ k doesn't matter}$$

Using the Euclidean Algorithm, solve the following until the remainder becomes 1

If the remainder equals 0, that means that the number does not have a multiplicative inverse mod m

$$\begin{array}{ll}
 \boxed{m} = k_1 \cdot \boxed{a} + \boxed{r_1} & \text{(first iteration)} \\
 \boxed{a} = k_2 \cdot \boxed{r_1} + \boxed{r_2} & \text{(second iteration)} \\
 \boxed{r_1} = k_3 \cdot \boxed{r_2} + \boxed{r_3} & \text{(third iteration)} \\
 \boxed{r_2} = k_4 \cdot \boxed{r_3} + 1 & \text{(remainder = 1)}
 \end{array}$$

For the factors, just use a calculator

Now using this work, solve for ever remainder, so in this case solve for

$$r_1, r_2, r_3, 1$$

Using the lines where they're the remainder

In order, we get

$$\begin{array}{ll}
 \boxed{r_1} = \boxed{m} - k_1 \cdot \boxed{a} & \text{(from first)} \\
 \boxed{r_2} = \boxed{a} - k_2 \cdot \boxed{r_1} & \text{(from second)} \\
 \boxed{r_3} = \boxed{r_1} - k_3 \cdot \boxed{r_2} & \text{(from third)} \\
 \boxed{1} = \boxed{r_2} - k_4 \cdot \boxed{r_3} & \text{(from last)}
 \end{array}$$

Now just work backwards and substitute each **divisor**, then combining like terms, but DO NOT actually multiply

Do this until you get to the first line with $\boxed{m}$. The inverse is the other factor with $\boxed{a}$

Example 7 (Euclidean Algorithm Practice)

Let's do some examples of the Euclidean Algorithm

Let's find the multiplicative inverse of $\boxed{37} \bmod \boxed{200}$

$$\begin{aligned}
 200 &= 5 \cdot 37 + 15 \\
 37 &= 2 \cdot 15 + 7 \\
 15 &= 2 \cdot 7 + 1
 \end{aligned}$$

Now that we found remainder = 1, let's solve for each remainder

$$\begin{aligned}
 15 &= 200 - 5 \cdot 37 \\
 7 &= 37 - 2 \cdot 15 \\
 1 &= 15 - 2 \cdot 7
 \end{aligned}$$

Now let's work backwards and substitute, starting with the bottom equation

$$\begin{aligned}
 1 &= 15 - 2(37 - 2 \cdot 15) \\
 1 &= 15 - 2 \cdot 37 + 4 \cdot 15 \\
 1 &= 5 \cdot 15 - 2 \cdot 37 \\
 1 &= 5(200 - 5 \cdot 37) - 2 \cdot 37 \\
 1 &= 5 \cdot 200 - 25 \cdot 37 - 2 \cdot 37 \\
 1 &= 5 \cdot 200 - 27 \cdot 37
 \end{aligned}$$

From this, we see that

$$37^{-1} = -27$$

But we don't like negatives, so

$$\begin{array}{rcl}
 37^{-1} & \equiv & -27 \pmod{200} \\
 +0 & \equiv & 200 \pmod{200} \\
 \hline
 37^{-1} & \equiv & 173 \pmod{200}
 \end{array}$$

1.3.2 Greatest Common Divisor (GCD)

Let's focus our attention to a the **gcd**

Definition 7 (Greatest Common Divisor, gcd)

The Greatest Common Divisor, or gcd, gets the greatest common divisor, or factor, between two numbers

$$a, b \in \mathbb{Z}, \text{ denoted as } \gcd(a, b)$$

Such that $\gcd(a, b)$ satisfies

1. $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$
2. $\gcd(a, b) = \max \{d \in \mathbb{Z}^+ \mid d \mid a \text{ and } d \mid b\}$

We can find the gcd by using the Euclidean Algorithm. Basically the GCD is the **last nonzero remainder**

So just do Euclidean Algorithm until the remainder is 0, don't stop at 1

Example 8 (GCD with Euclidean Algorithm)

Determine

$$\gcd(142856, 882)$$

$$\begin{aligned} 142856 &= 161 \cdot 882 + 854 \\ 882 &= 1 \cdot 854 + 28 \\ 854 &= 30 \cdot 28 + 14 \\ 28 &= 2 \cdot 14 + 0 \end{aligned}$$

Since we found remainder = 0, the GCD is the last nonzero remainder, which is **14** in this case

$$\gcd(142856, 882) = 14$$

Theorem 2 (Bézout's Theorem)

Basically

$$a, b \in \mathbb{Z}^+ \implies \exists x, y \in \mathbb{Z}, ax + by = \gcd(a, b)$$

Theorem 3

If $a \mid bc$ and $\gcd(a, c) = 1$ then $a \mid b$

Proof

Since $\gcd(a, c) = 1$ then by Bézout's

$$ax + cy = \gcd(a, c) = 1$$

Multiply both sides by b and we get

$$bax + bcy = b$$

Since $a \mid bc$, that means $a \mid bcy$, and $a \mid bax$, hence $a \mid (bax + bcy)$

$$\therefore a \mid b$$

Theorem 4

If $a, b, c \in \mathbb{Z}, m \in \mathbb{Z}^+$, with $\gcd(c, m) = 1$

Then $ac \equiv bc \pmod{m}$ implies $a \equiv b \pmod{m}$

Proof

$ac \equiv bc \pmod{m}$ implies $m \mid (ac - bc)$ or $m \mid (a - b)c$

By the previous theorem

$$m \mid (a - b)$$

Therefore

$$a \equiv b \pmod{m}, \text{ as desired}$$

Note

Division by an integer $c \pmod{m}$ works if and only if $\gcd(c, m) = 1$. You may need to use Euclidean Algorithm to find c^{-1}

1.3.3 Divisors

Let's look at how to find the number of divisors in a number n

Definition 8 (Prime Factorization)

Prime factorization is breaking up a number into prime factors. An example is

$$\begin{aligned}
 24 &= 6 \cdot 4 \\
 &= (3 \cdot 2)(2 \cdot 2) \\
 &= 3^1 \cdot 2^3
 \end{aligned}$$

The prime factors are 3 and 2.

In order to find the number of divisors, we must first break up n into its prime factors

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$$

p_k : Prime factor
 e_k : Occurences

The number of divisors is then

$$(e_1 + 1)(e_2 + 1) \dots (e_k + 1)$$

1.4 Section 4.4 - Linear Congruences

To think of linear congruences, let's start with an example

Example 9

A group of soldiers line up in rows of 8. However, there are 4 soldiers left over. When the soldiers line up in rows of 7, there are 5 soldiers left over. What is the least possible number of soldiers in the group?

Let x = number of soldiers

From the description, we know:

$$x \equiv 4 \pmod{8}$$

$$x \equiv 5 \pmod{7}$$

This also means that

$$\exists k_1 \in \mathbb{Z} \text{ such that } x = 8k_1 + 4$$

$$8k_1 + 4 \equiv 5 \pmod{7} \quad (\text{substitute } x)$$

$$8k_1 \equiv 1 \pmod{7}$$

$$k_1 \equiv 1 \pmod{7} \quad (\text{since } 8^{-1} \text{ is } 1)$$

Expanding that last equation, we get

$$\exists k_2 \in \mathbb{Z}, k_1 = 7k_2 + 1$$

Substituting k_1 back into the equation for x , we get

$$x = 8k_1 + 4 \quad (\text{original})$$

$$x = 8(7k_2 + 1) + 4 \quad (\text{substitute})$$

$$x = 56k_2 + 12$$

Since we're looking for the smallest number of soldiers, we can set $k_2 = 0$ and we get $x = 12$ as our final answer

This is actually an example of the **Chinese Remainder Theorem**

Definition 9 (Relatively Prime)

Two numbers $a, b \in \mathbb{Z}$ are **relatively prime** if

$$\gcd(a, b) = 1$$

Theorem 5 (Chinese Remainder Theorem)

Basically

Let $m_1, m_2, \dots, m_n \in \mathbb{Z}^+$ which are pairwise relatively prime (they're all relatively prime to each other) and $a_1, a_2, \dots, a_n \in \mathbb{Z}$, then the system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$\vdots$$

$$x \equiv a_n \pmod{m_n}$$

Has a unique solution that's mod $M = m_1 \cdot m_2 \cdot \dots \cdot m_2$

Example 10

Let's start with the system

$$x \equiv 14 \pmod{16}$$

$$x \equiv 21 \pmod{31}$$

$$x \equiv 12 \pmod{41}$$

We start by rewriting the first equation

$$x = 16k_1 + 14$$

And substituting it into the second equation

$$16k_1 + 14 \equiv 21 \pmod{31}$$

Now isolate k_1 . You need to find 16^{-1} , but thankfully we have some shortcuts

$$16k_1 \equiv 7 \pmod{31}$$

$$32k_1 \equiv 14 \pmod{31}$$

$$k_1 \equiv 14 \pmod{31}$$

(multiply 2)

$$(32 \equiv 1 \pmod{31})$$

Now that we have k_1 , let's expand it

$$k_1 = 31k_2 + 14$$

And plug it back into our equation of x

$$x = 16k_1 + 14$$

(original)

$$x = 16(31k_2 + 14) + 14$$

$$x = 16 \cdot 31k_2 + 16 \cdot 14 + 14$$

$$x = 496k_2 + 238$$

Now that we have a new equation for x involving k_2 , let's use our last equation

$$\begin{aligned}
 x &\equiv 12 \pmod{41} && \text{(last equation)} \\
 496k_2 + 238 &\equiv 12 \pmod{41} \\
 496k_2 &\equiv -226 \pmod{41} \\
 4k_2 &\equiv 20 \pmod{41} && \text{(reduce 496 and -226)}
 \end{aligned}$$

Now we gotta find $4^{-1} \pmod{41}$

Let's use the Euclidean Algorithm

$$\begin{aligned}
 41 &= 10 \cdot 4 + 1 \\
 1 &= 41 - 10 \cdot 4 \\
 4^{-1} &\equiv -10 \pmod{41} \\
 \boxed{4^{-1} &\equiv 31 \pmod{41}} && \text{(add 41)}
 \end{aligned}$$

Now that we have 4^{-1} , we solve

$$\begin{aligned}
 4k_2 &\equiv 20 \pmod{41} \\
 4^{-1} \cdot 4k_2 &\equiv 4^{-1} \cdot 20 \pmod{41} \\
 31 \cdot 4k_2 &\equiv 31 \cdot 20 \pmod{41} \\
 k_2 &\equiv 620 \pmod{41} \\
 \boxed{k_2 &\equiv 5 \pmod{41}}
 \end{aligned}$$

Now that we solved for k_2 , let's expand it one last time

$$k_2 = 41k_3 + 5$$

And plug it back into our equation for x involving k_2

$$x = 496k_2 + 238 \quad (\text{original})$$

$$x = 496(41k_3 + 5) + 238 \quad (\text{substituted})$$

$$\boxed{x = 20336k_3 + 2718} \quad (\text{simply})$$

Now that we simplified for the equation, we see that it represents

$$\boxed{x \equiv 2718 \pmod{20336}}$$

And you can see from the Chinese Remainder Theorem that our mod is actually the product of our moduli

$$M = 20336 = 16 \cdot 31 \cdot 41$$

That was... very long, so let's write out the steps:

1.4.1 Steps for Solving Linear Congruences

Given the system: (I'm just gonna use up to 3 for this)

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_3 \pmod{m_3}$$

The system has a **unique solution** if m_1, m_2, m_3 are pairwise coprime. If not, then it *could possibly* have a solution but not necessarily

1. Expand the first equation

$$x \equiv a_1 \pmod{m_1}$$

implies

$$\boxed{x = m_1k_1 + a_1}, k_1 \in \mathbb{Z}$$

2. Plug that into the second equation

$$\begin{aligned}x &\equiv a_2 \pmod{m_2} \\ m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2}\end{aligned}$$

And solve for k_1

$$\begin{aligned}m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \\ m_1 k_1 &\equiv a_2 - a_1 \pmod{m_2} \\ m_1^{-1} k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\ k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\ \boxed{k_1 \equiv c_1} &\pmod{m_2} \quad (c_1 \text{ is shorthand})\end{aligned}$$

Note

This isn't a formula, just steps with arbitrary variables. You'll need to find the inverse of m_1

And expand that last equation

$$k_1 \equiv c_1 \pmod{m_2}$$

implies

$$\boxed{k_1 = m_2 k_2 + c_1}, \quad k_2 \in \mathbb{Z}$$

3. Plug that into our equation for x

$$\begin{aligned}x &= m_1 k_1 + a_1 \\ \boxed{x &= m_1(m_2 k_2 + c_1) + a_1}\end{aligned}$$

Plug that into the last equation

$$m_1(m_2k_2 + c_1) + a_1 \equiv a_3 \pmod{m_3}$$

and solve for $k_2 \pmod{m_3}$

$$\vdots$$

$$\boxed{k_2 \equiv c_2 \pmod{m_3}}$$

Expand that equation

$$\boxed{k_2 = m_3k_3 + c_2}, k_3 \in \mathbb{Z}$$

4. Finally, plug that last equation into our SECOND equation for x

$$x = m_1(m_2k_2 + c_1) + a_1$$

$$x = m_1(m_2(m_3k_3 + c_2) + c_1) + a_1$$

$$\boxed{x = m_1m_2m_3k_3 + c_3}$$

Now we UN-expand this equation and see that

$$\boxed{x \equiv c_3} \pmod{m_1m_2m_3}$$

Basically, starting with the first equation

1. Expand first, solving for x
2. Plug x into second
3. Solve for k_1
4. Expand k_1
5. Plug k_1 back into first expanded of x (one with k_1)

6. Solve for k_2
7. Expand k_2
8. Plug k_2 back into second expanded of x (one with k_2)
9. Unexpand back into modulus

Just do these lines for the first equation, and these lines for each other equation

Let's do something else now

1.4.2 Factorials

1. $1! \pmod{2} = 1$
2. $2! \pmod{3} = 2$
3. $3! \pmod{4} = 2$
4. $4! \pmod{5} = 4$
5. $5! \pmod{6} = 0$, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{6} = 0$
6. $6! \pmod{7} = 6$
7. $7! \pmod{8} = 0$, $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{8} = 0$
8. $8! \pmod{9} = 0$, $8! = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{9} = 0$
9. $9! \pmod{10} = 0$, $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{10} = 0$
10. $10! \pmod{11} = 10$
11. $999! \pmod{1000} = 0$ since 1000 is composite

Notice that for all the ones that are not 0, the modulus is prime (except 4)

This brings us to our next theorems

1.4.3 Wilson's Theorem

Theorem 6 (Wilson's Theorem)

If p is a prime integer, then

$$(p-1)! \equiv p-1 \pmod{p}$$

or

$$(p-1)! \equiv -1 \pmod{p}$$

But for composite numbers, we have

Theorem 7 (Some Unnamed Theorem)

If n is a composite integer greater than 4,
then

$$(n-1)! \equiv 0 \pmod{n}$$

If $n = 4$, then

$$3! \equiv 2 \pmod{4}$$

1.4.4 Fermat's Little Theorem

There's examples that derive this, but I'm too lazy

Theorem 8 (Fermat's Little Theorem)

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof

Consider $S = \{1, 2, \dots, p-1\}, p \in \mathbb{Z}$

Let's define a function $f : S \rightarrow S, f(x) = ax \pmod{p}$

Example 11

Let $p = 7, a = 3$

Then $S = \{1, 2, 3, 4, 5, 6\}$ and $f(x) = 3x \pmod{7}$

x	$f(x)$
1	3
2	6
3	2
4	5
5	1
6	4

We see that f is a **bijection**

so we claim f is a **bijection**

Why?

Let $y \in S$

We must solve

$$ax \equiv y \pmod{p}$$

and since $\gcd(a, p) = 1$, then a^{-1} exists $\pmod{p}$

Therefore, $a^{-1}(ax) \equiv a^{-1}y \pmod{p}$

or

$$x \equiv a^{-1}y \pmod{p}$$

we conclude that f is **onto (surjective)**

For the injective part, suppose $f(x_1) = f(x_2)$

Then

$$ax_1 \equiv ax_2 \pmod{p}$$

Since

$$\gcd(a, p) = 1, \text{ then } a^{-1} \text{ exists } \pmod{p}$$

Therefore

$$a^{-1}(ax_1) \equiv a^{-1}(ax_2) \pmod{p}$$

or

$$x_1 \equiv x_2 \pmod{p}$$

We conclude f is **one to one (injective)**

Now let's consider the product $f(1) \cdot f(2) \cdot \dots \cdot f(p-1)$

$$f(1) \cdot f(2) \cdot \dots \cdot f(p-1) \equiv 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \pmod{p}$$

$$a \cdot 2a \cdot \dots \cdot (p-1)a \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

$$a^{p-1} [1 \cdot 2 \cdot \dots \cdot (p-1)] \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

By Wilson's Theorem we get

$$(p-1)! \equiv p-1 \pmod{p}$$

$$\begin{aligned}
a^{p-1}! &\equiv (p-1)! \pmod{p} \\
a^{p-1}(p-1) &\equiv p-1 \pmod{p} \\
a^{p-1} &\equiv 1 \pmod{p} \text{ as desired}
\end{aligned}$$

Let's do some examples

Example 12

Compute the following

$$\begin{aligned}
&9^{2026} \pmod{11} \\
&a : 9 \\
&p : 11
\end{aligned}$$

$$\gcd(9, 11) = 1$$

By Fermat's Little Theorem,

$$9^{10} \equiv 1 \pmod{11}$$

We can use this to break up 2026

$$\begin{aligned}
9^{2026} &\equiv 9^{202 \cdot 10 + 6} \pmod{11} \\
&\equiv 9^{202 \cdot 10} \cdot 9^6 \pmod{11} \\
&\equiv 9^6 \pmod{11}
\end{aligned}$$

Remember that we can subtract 11 from 9, so $9 \equiv -2 \pmod{11}$

This makes

$$\begin{aligned}
&\equiv 9^6 \pmod{11} \\
&\equiv (-2)^6 \pmod{11} \\
&\equiv 64 \pmod{11} \\
&\equiv 9 \pmod{11}
\end{aligned}$$

1.4.5 Extended Fermat's Little Theorem

There's also an extension of this

Theorem 9 (Extended Fermat's Little Theorem)

If $a, p, q \in \mathbb{Z}$ with

1. p and q are distinct primes
2. $\gcd(a, pq) = 1$

Then

$$a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$$

Example 13

Compute

$$1. \quad 13^{2026} \pmod{253}$$

First, let's check:

- (a) $253 = 11 \cdot 23$
 $p = 11, q = 23$, are distinct primes
- (b) $\gcd(13, 253) = 1$

This means we can use Extended Fermat's Little Theorem (EFLT)

$$\begin{aligned}
13^{(11-1)(23-1)} &\equiv 1 \pmod{253} \\
13^{10 \cdot 22} &\equiv 1 \pmod{253} \\
13^{220} &\equiv 1 \pmod{253}
\end{aligned}$$

Breaking up 2026,

$$2026 = 9 \cdot 220 + 46 \pmod{253}$$

This means

$$13^{2026} \equiv 13^{46} \pmod{253}$$

$$13 \pmod{253} = 13$$

$$\begin{aligned} 13^2 \pmod{253} &= 13^2 \\ &= 169 \end{aligned}$$

$$\begin{aligned} 13^4 \pmod{253} &= 169^2 \pmod{253} \\ &= 225 \end{aligned}$$

$$\begin{aligned} 13^8 \pmod{253} &= 225^2 \pmod{253} \\ &= 25 \end{aligned}$$

$$\begin{aligned} 13^{16} \pmod{253} &= 25^2 \pmod{253} \\ &= 119 \end{aligned}$$

$$\begin{aligned} 13^{32} \pmod{253} &= 119^2 \pmod{253} \\ &= 246 \end{aligned}$$

Now let's break up 46 into powers of 2

$$\begin{aligned} 13^{2026} &\equiv 13^{46} \\ &\equiv 13^{32} \cdot 13^8 \cdot 13^4 \cdot 13^2 \pmod{253} \\ &\equiv 246 \cdot 25 \cdot 225 \cdot 169 \pmod{253} \\ &\equiv 31 \end{aligned}$$

2.

$$24^{2026} \pmod{31}$$

(a) $p = 31$ is prime

(b) $\gcd(24, 31) = 1$

so FLT applies

$$24^{31-1} \equiv 1 \pmod{31}$$

$$24^{30} \equiv 1 \pmod{31}$$

$$2026 = 67 \cdot 30 + 16$$

$$\begin{aligned}
 24^{2026} &\equiv 24^{67 \cdot 30 + 16} \pmod{31} \\
 &\equiv (24^{30})^{67} \cdot 24^{16} \pmod{31} \\
 &\equiv 1^{67} \cdot 24^{16} \pmod{31} \\
 &\equiv 24^{16} \pmod{31} \\
 &\equiv 7 \pmod{31} \qquad \text{(brute forced)}
 \end{aligned}$$

1.4.6 An Implication of Fermat's Little Theorem

Definition 10 (Implication of FLT)

I'm not sure if this is called something but recall that Fermat's Little theorem states that

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

But actually, any multiple of $p - 1$ will make a equivalent to 1 mod p

We can rewrite this as

$$a^{k(p-1)} \equiv 1 \pmod{p}, k \in \mathbb{Z}^+$$

But this another way of saying this is

$$a^k \equiv 1 \pmod{p}, k \equiv 0 \pmod{p-1}$$

The second half just means that k is a multiple of $p - 1$

So TDLR, if a is raised to a multiple of $p - 1$, then it is equivalent to 1 mod p

This also implies that

$$a^x \equiv a^y \pmod{p} \text{ if } x \equiv y \pmod{p-1}$$

We'll see this being used later

1.6 Section 4.6 - Cryptography

Sorry in advance to anyone reading this since the professor never really gave a proper intro to this section and I'm too lazy right now to write an actual intro

But basically this section is about methods of encrypting data

Let's start with defining our messages

<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>	<u>E</u>
00	01	02	03	04
F	G	H	I	J
<u>05</u>	<u>06</u>	<u>07</u>	<u>08</u>	<u>09</u>
K	L	M	N	O
<u>10</u>	<u>11</u>	<u>12</u>	<u>13</u>	<u>14</u>
P	Q	R	S	T
<u>15</u>	<u>16</u>	<u>17</u>	<u>18</u>	<u>19</u>
U	V	W	X	Y
<u>20</u>	<u>21</u>	<u>22</u>	<u>23</u>	<u>24</u>
				Z
				25

Definition 11 (Encryption)

Given a message P (plain text), like **H E L L O**

using some method b to change the message into C (cyphertext)

Example 14

Let's use P from earlier, **H E L L O**, and get C by shifting the number value of each letter by +1

<u>H</u>	<u>E</u>	<u>L</u>	<u>L</u>	<u>O</u>
07	04	11	11	14
		↓ ₊₁		
<u>I</u>	<u>F</u>	<u>M</u>	<u>M</u>	<u>P</u>
08	05	12	12	15

or you could do

$$x \mapsto ax + b$$

This is called the **Caesar Cipher**

Definition 12 (Exponentiation Cycle)

Let's define this by doing an example

Example 15

Let's call P M for this example (M for message)

We need to choose a prime number bigger than 26 (since there are 26 letters)

Let's do $p = 43$

and choose an exponent

Let's do $e = 5$

Remember that our encrypted message is C

Let's just do the $\underbrace{\text{H}}_{07}$ in HELLO since I'm lazy

Let's start by **encrypting** our message.

This means that

$$p = 43$$

$$e = 5$$

$$M = 7$$

$$\begin{aligned} C &\equiv M^e \pmod{p} \\ C &\equiv 7^5 \pmod{43} \\ &\equiv 16,807 \pmod{43} \\ &\equiv 37 \pmod{43} \end{aligned}$$

Boom, so 37 is our encrypted message that the receiver has to decrypt.

Now let's look at how to **decrypt** the message

We KNOW

$$C \equiv M^e \pmod{p}$$

But we WANT

$$C^d \equiv M \pmod{p}$$

(trying to find d) because we need to find M

Plugging the KNOW (first equation) into the WANT (second equation), we get

$$\begin{aligned}
(M^e)^d &\equiv M \pmod{p} \\
M^{ed} &\equiv M \pmod{p} \\
M^{ed} \cdot M^{-1} &\equiv M \cdot M^{-1} \pmod{p} \\
M^{ed-1} &\equiv 1 \pmod{p}
\end{aligned}$$

Recall 1

From 4.4, we saw that

$$a^k \equiv 1 \pmod{p}, k \equiv 0 \pmod{p-1}$$

In this case, $k = ed - 1$, and we see that

$$\begin{aligned}
ed - 1 &\equiv 0 \pmod{p-1} \\
\boxed{ed &\equiv 1 \pmod{p-1}}
\end{aligned}$$

And since we know e and p , we can find d

$$\begin{aligned}
ed &\equiv 1 \pmod{p-1} \\
5d &\equiv 1 \pmod{42} \\
5d &\equiv 85 \pmod{42} && (\text{add } 2 \cdot 42) \\
5d &\equiv 5 \cdot 17 \pmod{42} \\
\boxed{d &\equiv 17 \pmod{42}}
\end{aligned}$$

Note

We used a shortcut way of finding the inverse but you'd typically use the Euclidean Algorithm

Recall 2

Also, note that we found what d is $\pmod{p-1}$ and not $\pmod{p}$. From 4.4 again, we can see

$$a^x \equiv a^y \pmod{p} \text{ if } x \equiv y \pmod{p-1}$$

There's a lot of algebra to prove that we can use this value of d even though it's a different mod, but I'm too lazy to prove it so just trust

Okay so now that we know that $d \equiv 17 \pmod{42}$

From the decrypter's standpoint, let's recap what know so far

$$p = 43$$

$$e = 5$$

$$d = 17$$

$$C = 37$$

We want to find M

Let's use the equation from earlier

$$C^d \equiv M \pmod{p}$$

So now we just solve

$$37^{17} \equiv M \pmod{43}$$

$$\begin{aligned} 37^{17} &\equiv (-6)^{17} \pmod{43} \\ &\equiv -(6)^{17} \pmod{43} \end{aligned}$$

$$6 \pmod{43} = 6$$

$$6^2 \pmod{43} = 36$$

$$\begin{aligned} 6^4 \pmod{43} &= 36^2 \\ &= 6 \end{aligned}$$

$$6^8 \pmod{43} = 36$$

$$6^{16} \pmod{43} = 6$$

$$\begin{aligned}
37^{17} &\equiv -(6)^{16} \cdot 6 \pmod{43} \\
&\equiv -6 \cdot 6 \pmod{43} \\
&\equiv -36 \pmod{43} \\
&\equiv 7 \pmod{43}
\end{aligned}$$

Which is what we expect!

Definition 13 (RSA Encryption)

An extension of what we just did is **RSA Encryption**, which uses **Extended Fermat's Little Theorem** instead of normal FLT

The way it works is you choose 2 distinct primes, p and q

$$p = 29, q = 47$$

$$\begin{aligned}
N &= pq = 29 \cdot 47 \\
&= 1363
\end{aligned}$$

And choose an exponent that is relatively prime of p and q

$$e = 5$$

Let's say our message is $\underbrace{H}_{07}$ again

$$M = 7$$

To encrypt, we do

$$C \equiv M^e \pmod{N}$$

Okay let's recap. We just went over 2 exponentiation cyphers

1. Given a prime p and exponent e that are relatively prime to each other (both public)

$$C \equiv M^e \pmod{p}$$

$$C^d \equiv M \pmod{p}$$

$$M^{ed} \equiv M \pmod{p}$$

$$M^{ed-1} \equiv 1 \pmod{p}$$

$$\implies ed \equiv 1 \pmod{p-1}$$

2. Given 2 distinct primes p, q and exponent e where $N = pq$ and e are both public

$$C \equiv M^e \pmod{N}$$

$$C^d \equiv M \pmod{N}$$

$$M^{ed} \equiv M \pmod{N}$$

$$M^{ed-1} \equiv 1 \pmod{N}$$

$$\implies ed \equiv 1 \pmod{(p-1)(q-1)}$$

Let's introduce another cypher, but let's start with an example

Example 16

Let's do HELLO again

$$\begin{array}{ccccc} \text{H} & \text{E} & \text{L} & \text{L} & \text{O} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 07 & 04 & 11 & 11 & 14 \end{array}$$

But let's put each letter into pairs (and let's add a random letter since HELLO is 5 letters long)

$\begin{array}{cc} \text{H} & \text{E} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 07 & 04 \end{array}$	$\begin{array}{cc} \text{L} & \text{L} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 11 & 11 \end{array}$	$\begin{array}{cc} \text{O} & \text{Z} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 14 & 25 \end{array}$
---	---	---

Let's now look at our first pair / block

H	E
07	04

$$\begin{aligned}
 M &= 0704 \\
 &= 704 \\
 p &= 43, q = 59 \\
 (p-1)(q-1) &= 42 \cdot 58 \\
 &= 2436 \\
 N &= pq = 2537 \\
 e &= 55
 \end{aligned}$$

Let's start by encrypting M , that is

$$\begin{aligned}
 C &\equiv M^e \pmod{N} \\
 C &\equiv 704^{55} \pmod{2537}
 \end{aligned}$$

Solving for C ...

Okay I'm too lazy to write it all out but just reduce 704^{55} into powers of 2 and you get

$$\begin{aligned}
 704^{55} &\equiv 637 \pmod{2537} \\
 C &\equiv \boxed{637} \pmod{2537}
 \end{aligned}$$

Now let's decrypt C , that is

$$M \equiv C^d \pmod{N}$$

To do this, we must solve

$$\begin{aligned}
 ed &\equiv 1 \pmod{(p-1)(q-1)} \\
 55d &\equiv 1 \pmod{2436}
 \end{aligned}$$

Now let's find 55^{-1}

We do this with the Euclidean Algorithm

$$2436 = 44 \cdot 55 + 16$$

$$55 = 3 \cdot 16 + 7$$

$$16 = 2 \cdot 7 + 2$$

$$7 = 3 \cdot 2 + 1$$

$$1 = 7 - 3 \cdot 2$$

$$2 = 16 - 2 \cdot 7$$

$$7 = 55 - 3 \cdot 16$$

$$16 = 2436 - 44 \cdot 55$$

$$1 = 7 - 3 \cdot (16 - 2 \cdot 7)$$

$$1 = 7 \cdot 7 - 3 \cdot 16$$

$$1 = 7(55 - 3 \cdot 16) - 3 \cdot 16$$

$$1 = 7 \cdot 55 - 24 \cdot 16$$

$$1 = 7 \cdot 55 - 24(2436 - 44 \cdot 55)$$

$$1 = 1063 \cdot 55 - 24 \cdot 2436$$

$$55^{-1} \equiv 1063 \pmod{2436}$$

$$d \equiv 1063 \pmod{2436}$$

$$M \equiv C^d \pmod{N}$$

$$M \equiv 637^{1063} \pmod{2537}$$

$$M \equiv 704 \pmod{2537}$$